

Fundamentals of Operating System Design

Module 3: Operating System Services

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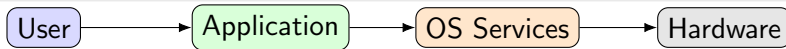
Learning Outcomes

- Explain operating system services.
- Classify common OS services.
- Relate OS services to real applications.
- Distinguish services from kernel components.

What are Operating System Services?

Definition

Operating system services are the facilities provided by the operating system to users and application programs for efficient and secure execution.



Major Operating System Services

- 1 Program Execution
- 2 I/O Operations
- 3 File System Manipulation
- 4 Communication
- 5 Error Detection
- 6 Resource Allocation
- 7 Accounting
- 8 Protection and Security

Program Execution

- Loads a program into memory.
- Creates a process.
- Allocates CPU time.
- Terminates the program after completion.

I/O Operations

- Keyboard input
- Display output
- Printer access
- Disk read/write
- Network communication

File System Services

- Create files
- Delete files
- Read and write files
- Rename files
- Manage directories

- Inter-process communication (IPC)
- Shared Memory
- Pipes
- Message Queues
- Network Communication

- User authentication
- Access permissions
- Process isolation
- Data protection
- Malware prevention

Summary

Service	Purpose
Program Execution	Run programs
I/O Operations	Access devices
File System	Manage files
Communication	Exchange information
Error Detection	Improve reliability
Resource Allocation	Share CPU and Memory
Accounting	Monitor usage
Protection	Secure the system

Quick Review

- ① What are OS services?
- ② List any four OS services.
- ③ Why is protection required?
- ④ Differentiate I/O service and file service.